7.1.2 Wave theory

Christian Huygens (1678) proposed the wave theory to explain the propagation of light through a medium. According to him, light is a disturbance from a source travels that as longitudinal mechanical waves through the ether medium (that was presumed to pervade all space) as mechanical wave requires a medium for its propagation. The wave theory could successfully explain phenomena of reflection, refraction, interference and diffraction of light.

Later, the existence of ether in all space was proved to be wrong. Hence, this theory could not explain the propagation of light through vacuum. The phenomenon of polarisation could not be explained by this theory as it is the property of only transverse waves.

7.1.3 Electromagnetic wave theory

Maxwell (1864) proved that light is an electromagnetic wave which is transverse in nature carrying electromagnetic energy. He could also show that no medium is necessary for the propagation of electromagnetic waves. All the phenomenon of light could be successfully explained by this theory.

Nevertheless, the interaction phenomenon of light with matter like photoelectric effect and Compton effect could not be explained by this theory.

7.1.4 Quantum theory

Albert Einstein (1905), endorsing the views of Max Plank (1900), was able to explain photoelectric effect (discussed in Unit 8) in which light interacts with matter as *photons* to eject the electrons. A photon is a discrete packet of energy. Each photon has energy E of,

$$E = h\nu \qquad (7.1)$$

Where, h is Plank's constant ($h = 6.625 \times 10^{-34}$ Js) and ν is frequency of electromagnetic wave.

As light has both wave as well as particle nature it is said to have dual nature. It is concluded that light propagates as a wave and interacts with matter as a particle.

7.2 WAVE NATURE OF LIGHT

Light is a transverse, electromagnetic wave. The wave nature of light was first demonstrated through experiments like interference and diffraction. The transverse nature of light is demonstrated in polarization. Like all electromagnetic waves, light can travel through vacuum.

7.2.1 Wave optics

Wave optics deals with the wave characteristics of light. Even the law of reflection and refraction are proved only with the help of wave optics. Though light propagates as a wave, its direction of propagation is still represented as a ray.

A good example for wave propagation is the spreading of circular ripples on the surface of still water from a point where a stone is dropped. The molecules (or) particles of water at a point are moving only up and down (oscillate) when a ripple passes through that point. All these particles on the circular ripple are in the same phase of vibration as they are all at the same distance from the center. The ripple represents a wavefront as shown in Figure 7.1(a). **A *wavefront* is the locus of points which are in the same state (or) phase of vibration.**

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